

SEAT No. _____

[45]

SARDAR PATEL UNIVERSITY

M.Sc.(IT) (SEMESTER - II) EXAMINATION

PS02CINT52: Software Engineering

Wednesday, 9th April, 2025

Total Number of pages : 2



Time: 10:00 a.m. to 01:00 p.m.

Max. Marks: 70

1.(A) Select the most appropriate option for each question :

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- (i) The _____ process model divides the activities into four major quadrants.
(A) Waterfall (B) Spiral (C) Prototyping (D) Iterative
- (ii) Which system helps in routine, day-to-day business operations?
(A) DSS (B) MIS (C) ES (D) TPS
- (iii) UML stands for
(A) Universal Modeling Language (B) Unified Modeling Language
(C) Uniform Modeling Language (D) None of these
- (iv) The document that contains all functional and non-functional requirements of a system is called
(A) Software Design Document (B) Project Charter
(C) Software Requirements Specification (SRS) (D) None of these
- (v) In a use-case diagram, each use-case is represented by _____.
(A) an ellipse (B) a circle (C) a rectangle (D) a diamond
- (vi) _____ of a module refers to the relationship among elements of the same module.
(A) Cohesion (B) Coupling (C) Complexity (D) Efficiency
- (vii) _____ is typically done by the programmer of a module.
(A) Unit testing (B) Beta testing (C) Black-box testing (D) None of these
- (viii) In a class diagram, private attributes of a class are shown with a _____ sign in front of the attribute name.
(A) Plus(+) (B) minus (-) (C) asterisk(*) (D) hash(#)

2. Answer the following questions (ANY SEVEN):

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- (i) Define : *software engineering*. What are the *goals* of software engineering ?
- (ii) Differentiate : *open* and *closed systems*.
- (iii) What are the *advantages* of a *modular design* ?
- (iv) Define : Work Breakdown Structure (WBS). State its importance.
- (v) Which *criteria* can be used for determining the *quality* of *software design* ?
- (vi) Differentiate : *unit testing* vs *integration testing*.
- (vii) Explain the concept of *information hiding* in object-oriented design.
- (viii) List *only the names* of various types of *UML diagrams*.
- (ix) How *classes* are represented in a *class diagram* ? What do you mean by an *aggregation* relationship between classes ?

- 3.(A) Describe the *waterfall model* giving its limitations. 6
- 3.(B) What is *agile software development*? Explain its principles and benefits. How is it different from traditional software development models? 6
- OR
- 3.(B) Describe the *spiral model* giving its advantages and disadvantages. 6
- 4.(A) Explain the classical *Systems Development Life Cycle (SDLC)* method for developing information systems. 6
- 4.(B) What is an SRS? Describe the *characteristics* of SRS. 6
- OR
- 4.(B) Describe the major issues in *software project planning*. 6
- 5.(A) Describe the main *characteristics* and *advantages* offered by an *object-oriented design* of a system. 6
- 5.(B) Distinguish between module *coupling* and *cohesion*. 6
- OR
- 5.(B) Explain the *use-case diagrams* giving any example. 6
- 6.(A) Write a short note on *coding standards and coding guidelines*. 6
- 6.(B) Explain any six examples of the *common coding errors* and associated suggestions. 6
- OR
- 6.(B) Distinguish between *black-box testing* and *white-box testing*. 6

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